



FOREST VERIFICATION REPORT

ANALYZED ZONE INFORMATION

Latitude	40.9129	Longitude	39.7482
Zone size (km)	10	Analysis date	2026-09-30

75

FOREST INTEGRITY SCORE / 100 Healthy Forest

DETAILED RESULTS

Indicator	Value	Interpretation
Current vegetation (NDVI)	0.725	Excellent
NDVI Trend 3 years (Sentinel-2)	+0.001	Stable
10-year trend (Landsat 8/9)	+0.396	Improving
40-year trend (Landsat 1984)	+0.409	Improving
Forest Status (FAO)	Forest recovery	Recuperation
Annual NDVI rate	+1.02% / year	Amelioration
NDVI change 5 years	-0.002	Stable
Leaf type	Broadleaf (Deciduous/Evergreen)	Flat leaves, high photosynthesis
Developmental stage	Mature forest (60-120 years)	Well established, dense canopy
Above-ground biomass	~91.6 t/ha	Medium biomass (50-150 t/ha)
Canopy Height (GEDI+S2 10m)	10.2m max / 4.5m median	Tree height from LiDAR (ISS) fused with Sentinel-2 at 10m resolution
Carbon Assessment (IPCC)	1,562,161 t CO ₂	CO ₂ sequestered in this forest zone — IPCC methodology (AGB+BGB × 0.47 × 3.67)
CO ₂ per hectare	198.9 t CO ₂ /ha	Carbon density per hectare
Zone area	7,854 ha	Analysis zone area in hectares
Carbon credit value	EUR 23,432,415	Estimated value at EUR 15/tonne CO ₂ (voluntary carbon market)
REDD+ MRV Monitoring	Compliant	This report provides satellite-based Measurement, Reporting and Verification (MRV) data for the analyzed forest zone, consistent with REDD+ monitoring requirements. CO ₂ sequestration calculated using IPCC Tier 1 methodology from NASA GEDI LiDAR biomass data.

Methodology	IPCC 2006/2019 — Tier 1	Above-ground biomass measured via NASA GEDI LiDAR (25m, ISS). Carbon fraction: 0.47. CO2 conversion factor: 3.67. Below-ground biomass ratio: 0.26 (IPCC root-to-shoot). Source: IPCC Guidelines for National GHG Inventories (2019).
Carbon credit potential	EUR 1,952,701 / year	Estimated annual value based on maintained forest preservation. For official REDD+ credit issuance under VCS or Gold Standard, independent third-party verification is required.
GEDI Tree Cover	27.4%	Percentage of the zone covered by trees according to NASA GEDI
AI Forest Type	Mixed Forest	AI-based classification
Deforestation Risk (AI)	Moderate	Moderate risk
Main cause (AI)	Seasonal drought stress combined with possible selective logging or natural canopy gaps	
Degradation signs (AI)	Negative NDWI (-0.673) indicates water stress; moderate NBR (0.426) suggests some canopy disturbance	
Recovery signs (AI)	High NDVI (0.729) and elevated EVI (1.973) indicate active photosynthetic activity and biomass accumulation	
AI Recommendation	Monitor moisture availability and implement water conservation practices; conduct field survey to assess logging intensity and plan restoration if needed.	
Sentinel-1 Radar (VV)	-8.223 dB	Dense forest (radar)
Forest cover	97.2%	Excellent
Fire pixels detected (FIRMS/year)	99 pixels	~25 events estimated
Overall trend	stable	
Satellites used	- ESA Sentinel-2 - NASA FIRMS - ESA Sentinel-1 - NASA Landsat 8/9 - NASA Landsat 4/5 - NASA GEDI LiDAR	
Deforestation Alert	STABLE — No significant change	
Area change (3 years)	+5.9 ha	+0.05%
Annual rate (area/yr)	+2 ha/yr	Recovery
Area change (10 years)	+12462.1 ha	+3548.95%
Annual rate 10yr (area)	+1246.2 ha/yr	Recovery

SCORE BREAKDOWN

Component	Points	Maximum
Current vegetation (NDVI)	25	25
NDVI trend (3 years)	15	25
Forest cover	25	25
Sentinel-1 Radar (VV)	10	15
Fire score (0=many fires, 10=none)	0	10
TOTAL	75	100

FIRE DETECTION METHODOLOGY: Fire pixels are detected by NASA FIRMS using VIIRS (375m) and MODIS (1km) thermal anomaly sensors. Raw pixel count represents thermal detections over 12 months in the analysis zone. Estimated fire events are calculated using NASA clustering methodology (~4 pixels per fire event average). This is a statistical approximation. For precise fire event mapping, consult NASA FIRMS directly at firms.modaps.eosdis.nasa.gov

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